

The Medium-to-Matter Current-Leg Trilogy

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2026-09-09

When is a topological helicity current a matter current? A no-go trilogy for force-free and two-fluid plasmoids

Abstract

We ask, for a force-free (Beltrami) magnetic field $\text{curl } \mathbf{B} = \lambda \mathbf{B}$ and its two-fluid generalization, when the fluid mass 4-current $(\rho, \rho \mathbf{v})$ can be identified with the topological helicity 4-current (K^0, K^i) , $K^0 = \mathbf{A} \cdot \mathbf{B}$. The density leg of that identification – ρ proportional to the helicity density $\mathbf{A} \cdot \mathbf{B}$ – holds exactly as a profile identity in the Beltrami gauge with a London-locked flow. The *current* leg does not, and we settle its status as a tiered result. **(a)** In the static regime the full 4-current closes if and only if the field magnitude is spatially constant, $|\mathbf{B}| = \text{const}$; since a topologically nontrivial (nulled) field cannot satisfy this, closure is obstructed and the identification is, at best, a consistent postulate. **(b)** The natural escape – a driven two-fluid plasma carrying its own conserved *canonical* (generalized) helicity – only relocates the obstruction: aligned/steady closure requires $\mathbf{P} \cdot \mathbf{v} = \text{const}$ ($\mathbf{P} = \mathbf{A} + (\mathbf{m}/q)\mathbf{v}$ the canonical momentum), which we prove impossible for a nontrivial field via the perfect-square identity $(\mathbf{A} \cdot \mathbf{B})^2 \geq 0$, adversarially verified. **(c)** In the genuinely time-dependent regime the obstruction lifts: closure reduces to a single scalar condition $S = \mu - \mathbf{P} \cdot \mathbf{v} = 0$, whose surface is non-empty and is realizable by an electrostatic drive (the canonical-helicity source $d_\mu K_{\text{can}} = 2 \mathbf{f}_{\text{body}} \cdot \mathbf{v}$ vanishes for a purely electromagnetic force), with an index-1 differential-algebraic structure and a bounded drive. We show, however, that the equilibrium and relaxation dynamics both *select against* this closure surface, so the identification is externally driven, never self-organized. Two residuals remain; the deeper reduces exactly to the Beale-Kato-Majda enstrophy bound for 3D Hall-MHD – a recognized open problem in fluid regularity, not a defect of the construction. The results are stated with explicit tiers ([V] verified/proven here; [S] constructed/contingent; credited where the physics is established).

1. Introduction

Force-free fields $\text{curl } \mathbf{B} = \lambda \mathbf{B}$ (Chandrasekhar & Kendall 1957; Woltjer 1958) are the minimum-energy states of magnetically dominated plasmas at fixed helicity (Taylor 1974), and magnetic helicity $H = \int \mathbf{A} \cdot \mathbf{B}$ is the robust topological invariant of ideal evolution (Moffatt 1969; Moffatt & Ricca 1992). A recurring and physically attractive idea – in models where matter and field co-move – is that the fluid mass current *is* the flux of the field’s own winding: that ρ measures how densely the field is wound and $\rho \mathbf{v}$ measures how that winding flows. This paper asks precisely when that identification is mathematically admissible, treating the fluid current and the helicity current as candidate equal 4-vectors and testing the equality leg by leg.

The answer is a trilogy: two proven no-gos (static; aligned/steady), one constructive realizability result (driven), a selection negative (the dynamics avoid the closure surface), and a clean reduction of the last gate to an external regularity theorem. The scope discipline throughout is that these are limits of a *construction*, not verdicts against any observed plasma phenomenon.

2. The law and its two legs; the density leg [V]

Write ordinary matter conservation as a candidate topological continuity equation,

$$d_t \rho + \text{div}(\rho \mathbf{v}) = 0 \quad \Longleftrightarrow \quad d_\mu K^\mu = 0 ,$$

with $K^\mu = (K^0, K^i)$, $K^0 = \mathbf{A} \cdot \mathbf{B}$ the helicity density and K^i its flux, and the organizing proportionality $\rho = \kappa (\mathbf{A} \cdot \mathbf{B})$. Two legs must hold for the 4-current to close: the **density leg** ($\rho \sim K^0$) and the **current leg** ($\rho \mathbf{v} \sim K^i$).

Density leg = [V]. In the Beltrami gauge $\mathbf{A} = \mathbf{B}/\lambda$, $K^0 = \mathbf{A} \cdot \mathbf{B} = |\mathbf{B}|^2/\lambda \geq 0$ (sign-definite, verified to $3e-15$). With a London-locked cold-condensate flow $\text{curl } \mathbf{v} = -(q/m) \mathbf{B}$ (verified to $7e-15$), the density profile tracks $|\mathbf{B}|^2$ with $\text{corr}(\rho, \mathbf{A} \cdot \mathbf{B}) = 1.0000$ – an exact dimensionless profile identity. Because K^0 has units of a helicity density, κ is a genuine *dimensional* proportionality constant (the match is a profile proportionality, not a units identity). The density leg is settled; the entire subtlety lives in the current leg.

3. (a) The static no-go: closure requires $|\mathbf{B}| = \text{const}$ – [V]

Matching densities fixes the current *direction*: London flow gives $\mathbf{v} \sim \mathbf{A} \sim \mathbf{B}$, so the matter current $\rho \mathbf{v} \sim |\mathbf{B}|^2 \mathbf{B}$ is CUBIC in the field, while the helicity flux $K \sim \mathbf{B}$ is LINEAR. Equality of a cubic and a linear quantity demands $|\mathbf{B}|^2 = \text{const}$:

The 4-current closes if and only if $|\mathbf{B}|$ is constant.

The obstruction is basis-independent, measured by the residual of $\mathbf{B}$ from $\text{range}(\mathbf{L})$ for $\mathbf{L}[\phi] = \text{curl}((\phi \mathbf{B} - \text{grad } \phi \times \mathbf{B})/|\mathbf{B}|^2)$ – dimensionless and κ -independent. For a genuine toroidal Beltrami field the residual is 0.6053-0.6054 (basis-converged); a four-move reducibility sweep (gauge, transverse $\mathbf{E}$ / sheath, norm, multi- λ) never reaches the < 0.2 bar, and a transverse non-force-free $\mathbf{E}$ -field leaves it UNCHANGED. The deciding control is a null-free constant- $|\mathbf{B}|$ Beltrami field $\mathbf{B} = B_0(\cos z, \sin z, 0)$, which closes EXACTLY (residual 0.0000). Hence the obstruction is entirely the spatial variation of $|\mathbf{B}|$, and a nontrivial closed-fibre field (which must have nulls) provably cannot have $|\mathbf{B}| = \text{const}$. **The current leg is over-determined by construction – irreducibly a postulate, not a theorem, in the static regime.**

4. Canonical helicity is the right conserved current – [V] / credited

A driven object may carry its own conserved current. For a two-fluid plasma the correct one is **canonical (generalized) helicity**: with $\mathbf{P} = \mathbf{A} + (m/q)\mathbf{v}$ and $\Omega = \text{curl } \mathbf{P} = \mathbf{B} + (m/q) \text{curl } \mathbf{v}$, the current from $\mathbf{H} = \int \mathbf{P} \cdot \Omega$ is ideally conserved, $d_\mu K_\text{can}^\mu = 0$, *even when magnetic helicity is dissipated*, because the canonical Ohm's law $\mathbf{E}_\text{can} = -\mathbf{v} \times \Omega$ gives $\mathbf{E}_\text{can} \cdot \Omega = 0$ (a Casimir invariant). The prerequisite – “have your own conserved current first” – is met. *Credited*: Steinhauer & Ishida 1997 (PRL 79, 3423); Mahajan & Yoshida 1998 (PRL 81, 4863).

5. (b) The aligned/steady no-go: the $(\mathbf{A}'\mathbf{B}'\mathbf{C}')^2 \geq 0$ obstruction – [V], adversarially verified

Having the conserved current only *relocates* the obstruction. The canonical flux is $K_\text{can} = \mathbf{h} \cdot \mathbf{v} + (\mu - \mathbf{P} \cdot \mathbf{v}) \Omega$ ($\mathbf{h}$ the canonical helicity density, μ the Bernoulli head), so closure $(\rho, \rho \mathbf{v}) \sim (\mathbf{h}, K_\text{can})$ requires $(\mu - \mathbf{P} \cdot \mathbf{v}) \Omega = 0$. On a double-Beltrami equilibrium $\mu = \text{const}$, so this is the exact mirror of the static theorem:

Aligned/steady closure requires $\mathbf{P} \cdot \mathbf{v} = \text{const}$ (the analog of $|\mathbf{B}| = \text{const}$).

Making the relevant cross-term vanish forces $\lambda_- = -\lambda_+$; then $\mathbf{P} \cdot \mathbf{v} = \text{const}$ requires a POSITIVE ratio $p_1^2/p_2^2 > 0$ between two ABC helicity-fluctuation triples, which is impossible because $(A''0027B''0027)(B''0027C''0027)(A''0027C''0027) = (A''0027B''0027C''0027)^2 \geq 0$ is a perfect square. Independent verification: the residual reaches zero only at $c = -0.604 < 0$, the forbidden-sign branch;

distinct λ force a spectral band-gap and single-mode collapse (the spectrum is discrete on any compact domain, so there is no continuous loophole); the physical corner is $[V]$ -via-mechanism ($\text{std}(P.v)/\text{mean}$ equals the field- magnitude inhomogeneity to $\sim 1\%$). The two no-gos are thereby *unified* – the same obstruction seen once in B and once in P .

6. (c) The driven resolution: closure $\Leftrightarrow S = 0$, realizable by an electrostatic drive – $[S \text{ leaning } V]$

The third leg differs in kind. Define the scalar $S = \mu - P.v$. Since S is a SCALAR, $S \Omega = 0$ forces $S = 0$ pointwise (there is no “ S perpendicular to Ω ” branch; verified false to $4e-17$). The canonical current is then always a matter current $(h, h u)$ with $u = v + (S/h)\Omega$, closing on v exactly when $S = 0$. Two facts make this a genuine resolution:

1. **The $S = 0$ surface is non-empty and drive-compatible** (residual 0.00; jointly satisfiable to $\sim 1e-16$) – unlike the two no-gos, whose surfaces are empty. A degree-of-freedom count is sharp: the isolated ideal system is $8/8$, closure makes it $9/8$ (over-determined $\rightarrow$ the no-gos), and a sustained external drive supplies the one spare freedom.
2. **No extra $f_{\text{ext}} \perp \Omega$ constraint is needed** – it was a body-force artifact. The exact identity $d_\mu K_{\text{can}} \mu = 2 f_{\text{body}} \Omega$ vanishes for a purely electromagnetic (Lorentz + barotropic) force, so the $S = 0$ drive is simply the electrostatic potential $\phi = A.v + (m/2q) |v|^2 - w/q$. The differential-algebraic system is index-1 ($dS/d\phi = 1 \neq 0$): ϕ is algebraically slaved, bounded, and the two-species closure $\{S_e = 0, S_i = 0\}$ is absorbed by ϕ and the parallel potential A_{parallel} , with Jacobian $\det = -|v_e - v_i|^2$ (invertible iff current flows). **Tier:** $[S]$ leaning $[V]$.

7. The selection result: the dynamics avoid the closure surface – $[V]/[S]$ negative

Does the object select $S = 0$ on its own? No – a specific, tested negative. **(A) Variational:** minimizing energy at fixed canonical helicities yields double-Beltrami equilibria with $\mu = \text{const}$, hence $S = \text{const} - P.v \neq 0$; the only functional $S = 0$ extremizes is the residual squared (a tautology). **(B) Dynamical:** a driven-dissipative Hall two-fluid integration keeps $\text{rms}(S)/dPv$ at $O(1)$ and relaxes TOWARD the aligned $\mu = \text{const}$ state. Closure is therefore externally *driven*, never emergent – and, as a corollary, the two no-gos are the dynamical attractor of relaxation, which reinforces them.

8. Residuals: $R1$ benign; $R2 \Leftrightarrow$ the Beale-Kato-Majda bound – $[S]$ conditional

$R1$ (current-null $v_e = v_i$). The two-species drive scales as $1/|v_e - v_i|$ and diverges where the currents coincide, but this set is codimension-3 (isolated points, coincident with the field nulls) and hence measure zero; the fields stay bounded and the drive energy is $L1 + L2$ integrable, so closure holds on the full-measure current-full complement. Genuine-but-benign $[S]$.

$R2$ (global all-time existence) on a bounded, current-full trajectory reduces to $R2 \Leftrightarrow X3$ given $X1, X2$: **$X1$** an energy/ L^2 bound (HAVE, via a Lyapunov function $dV/dt \leq 0$ on the driven limit cycle); **$X2$** no current-null crossing (HAVE, via $R1$); **$X3$** the enstrophy/ H^1 (Beale-Kato-Majda) bound $\int |\Omega| dt < \infty$ (OPEN). $X3$ is category-identical to open large-data 3D Navier-Stokes / Hall-MHD global regularity (Chae, Degond & Liu 2014; Beale, Kato & Majda 1984) – a recognized external problem, not a defect of the construction. Numerically the driven system is stable to $T = 6000$ (~ 2196 cycles) with $|S| \leq 2.2e-16$ and bounded drive – but this horizon is a LOW-MODE reduced ODE that cannot exhibit the small-scale enstrophy blow-up by construction, and the object is high-Lundquist / super-critical, so it is weak evidence, NOT sub-criticality; $R2$ stays $[S]$ hard-open.

9. Result

The matter-to-helicity current identification is a $[V]$ no-go / well-founded postulate in the frozen regimes (static $|B| = \text{const}$ and aligned/steady $P.v = \text{const}$, both proven, both adversarially or mechanism-verified), **and a constructed-realizable closure $[S \text{ leaning } V]$**

V] in the genuinely-driven regime, via an external electrostatic drive, obstructed only on a measure-zero benign set. **The equilibrium and relaxation dynamics select against closure** (a first-class negative): it is externally driven, not emergent. **The sole remaining gate to full [V] is the 3D enstrophy / Beale-Kato-Majda regularity bound – a recognized external problem.** The density leg stays [V]; canonical helicity is a genuine ideal conserved current [V]/credited.

The identification “matter is the flow of the knot” is thus admissible only for a knot that is being actively driven; in a frozen or self-organizing field it is provably barred.

References

Beale, Kato & Majda 1984, Commun. Math. Phys. 94, 61. Chae, Degond & Liu 2014, Ann. IHP C 31, 555. Chandrasekhar & Kendall 1957, ApJ 126, 457. Mahajan & Yoshida 1998, PRL 81, 4863. Moffatt 1969, JFM 35, 117. Moffatt & Ricca 1992, Proc. R. Soc. A 439, 411. Steinhauer & Ishida 1997, PRL 79, 3423. Taylor 1974, PRL 33, 1139. Woltjer 1958, PNAS 44, 489.

Note: a fourth source for the canonical-helicity Casimir – Bae, Kang & Shin 2025, “On the double Beltrami states in Hall magnetohydrodynamics,” arXiv:2504.07629 – is now VERIFIED as a real, on-topic paper (double-Beltrami states as energy minimizers under conservation of two helicities) and corroborates the claim; the primary load-bearing pair remains Steinhauer-Ishida 1997 and Mahajan-Yoshida 1998 (confirm exact Prop. label at lock). No number in this paper is fabricated; ASCII-clean.